An Embedded System Based on Wilson-Cowan Oscillators for Multi-Gait Locomotion in Legged Robots

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***Abstract*—Legged robot locomotion is often inspired by biological systems, which move efficiently through a wide range of harsh environments. Reproducing this efficiency in artificial systems remains difficult, mainly due to the mechanical complexity and synchronization of the legs, stability requirements, and power consumption. We propose a low-cost, fully reconfigurable embedded system for real-time multi-gait locomotion in legged robots. The system implements a Central Pattern Generator (CPG) composed of four coupled Wilson-Cowan neural oscillators, one per leg, running on a single Arduino-based microcontroller. By switching between four connecting weight matrices, the same neural architecture generates the four canonical quadruped gaits – walk, trot, pace and gallop – and streams the resulting joint commands, via inverse kinematics, to a Lynxmotion SQ3U quadruped robot in real time. The system was implemented and validated in hardware using a Simulink-Arduino co-simulation, oscilloscope measurements, and physical walking trials on the robot, confirming that the generated rhythmic patterns preserve the phase relationships expected for each gait and produce coordinated locomotion. These results support CPG-based controllers as a lightweight and reconfigurable alternative to trajectory-based control for resource-constrained legged robots.**

***Index Terms*—central pattern generator, Wilson-Cowan oscillators, legged robots, embedded systems, multi-gait locomotion**

I. Introduction

Locomotion in both vertebrates and invertebrates is driven by complex neural circuits known as Central Pattern Generators (CPGs), which integrate cell-to-cell synaptic connections and the interactions between excitatory and inhibitory neurons to produce rhythmic, patterned outputs such as walking, trotting and galloping [1]–[4]. Biologically, it is well established that locomotion in animals is controlled by neural networks located in the spinal cord, which govern gait generation and transition [5], [6]. Theoretically, various oscillator-based models have been proposed to replicate this neural behavior [7]–[9], and among them, the Wilson-Cowan model of coupled excitatory and inhibitory neural populations has been particularly influential for designing CPG controllers for quadruped and hexapod platforms alike [14]. Despite significant progress in theoretical modeling, a major limitation is that many of these models have only been validated in software simulations, overlooking key challenges such as mechanical complexity, limb synchronization, stability, and power consumption in real-world robotic applications [10], [11].

Among the oscillator formulations available for CPG design, the two most widely used biologically-grounded alternatives are the Matsuoka oscillator [17], based on mutually-inhibiting neurons with adaptation, and the Wilson-Cowan model [12], [13], based on reciprocally-coupled excitatory and inhibitory neural populations. We adopt the Wilson-Cowan formulation for three reasons relevant to a low-cost embedded implementation. First, unlike the Matsuoka oscillator, whose sustained oscillations rely on an explicit adaptation (fatigue) state per neuron, the Wilson-Cowan pair (u, v) oscillates from the excitatory-inhibitory feedback loop alone, which reduces the number of state variables and parameters that must be integrated in real time on a resource-constrained microcontroller. Second, the network-level extension in (2) lets the inter-leg phase relationship of an arbitrary number of oscillators be set entirely through the sign and magnitude

of the connecting weight matrix W (Section II), which is what enables gait switching as a simple matrix replacement, as shown in [14]. Third, the bounded sigmoidal nonlinearity f_μ keeps u_i, v_i within a fixed range regardless of W, which simplifies the fixed-point arithmetic and normalization used on the embedded platform. Alternative formulations, such as Hopf oscillators [9] or the Matsuoka model, are equally capable of generating multi-gait rhythms and are not excluded by our approach; the comparison above motivates our specific choice rather than establishing the general superiority of the Wilson-Cowan model.

To enable real-time, autonomous navigation in legged robots, bio-inspired robotics requires an embedded system that integrates locomotion control, sensing, and decision-making. At the heart of this system is a processing unit – such as an FPGA, microprocessor, microcontroller, or single-board computer – that orchestrates the robot's movements and interactions with its environment [15]. Although CPG-based locomotion has been extensively explored in simulation, there is comparatively little research on hardware implementations, where limited memory and processing power, communication latency, and the need to keep the system inexpensive and easy to reconfigure come into play.

In this work, we present a novel embedded system based on an Arduino board that autonomously generates multi-gait locomotion patterns – walk, trot, pace and gallop – for a legged robot. Unlike FPGA- or GPU-based solutions, which typically require higher cost and more specialized development tools, our approach shows that a full multi-gait CPG controller can run in real time on a low-cost, widely available microcontroller, with gait switching handled simply by replacing the oscillators' connecting weight matrix rather than by re-programming the system.

Table I situates this work with respect to prior Wilson-Cowan-based CPG implementations. Li *et al.* [14] validated their four-oscillator model only in simulation, without hardware deployment. Nakada *et al.* [16] implemented a Wilson-Cowan oscillator network directly in analog CMOS circuitry for a quadruped walking robot; while this achieves low power consumption, gait switching requires re-fabricating or re-wiring the analog circuit rather than a software change. Our own prior work [15] implemented a neuromorphic locomotion system on an FPGA, which offers high parallelism and timing precision but at a higher hardware cost and design complexity than a general-purpose microcontroller. In contrast, the present system trades some of the raw timing precision of an FPGA for low cost, wide availability, and software-level reconfigurability, making it attractive for educational and rapid-prototyping settings.

The remainder of this paper is organized as follows: Section II describes the Wilson-Cowan CPG model and the weight-matrix design for the four gaits; Section III describes the hardware platform; Section IV reports the real-time implementation results; and Section V concludes the paper.

TABLE I: Comparison with related Wilson-Cowan CPG implementations

Work	**Platform**	**HW validated**	**Gait switching**
Li *et al.* [14]	Simulation only	No	Software (sim.)
Nakada *et al.* [16]	Analog CMOS	Yes	Circuit re-design
Rostro-Gonzalez [15]	FPGA	Yes	Software (HDL)
This work	Arduino (MCU)	Yes	Software (weight matrix)

II. Wilson-Cowan CPG Model

The dynamics of a continuous-time Wilson-Cowan neural oscillator are described by

$$\begin{aligned} T_u \frac{du}{dt} &= -u + f_\mu\left(au - bv + S_u\right); \\ T_v \frac{dv}{dt} &= -v + f_\mu\left(cu - dv + S_v\right); \\ f_\mu(x) &= \tanh(\mu x); \end{aligned} \tag{1}$$

where u and v express the activity of a group of excitatory and inhibitory neurons, parameters a through d denote the coupling strength between the groups of neurons (b: strength of coupling from v to u; c: strength of coupling from u to v), S_u and S_v are external inputs, T_u and T_v are time constants, and μ is the gain of the sigmoid transfer function $f_\mu(x)$.

Our CPG network is composed of four such interconnected oscillators (one per leg), forming a holosymmetric structure that supports stable, adaptive locomotion. With connecting weight matrix $W \in \mathbb{R}^{4\times4}$, the network is modeled as

$$\begin{aligned} T_{u_i} \frac{du_i}{dt} &= -u_i + f_\mu\Big(au_i - bv_i + \textstyle\sum_{j=1}^{n} W_{ij}u_j + S_{u_i}\Big); \\ T_{v_i} \frac{dv_i}{dt} &= -v_i + f_\mu\Big(cu_i - dv_i + \textstyle\sum_{j=1}^{n} W_{ij}v_j + S_{v_i}\Big); \\ y^i_{out} &= u_i - v_i; \quad i,j = 1,2,3,4. \end{aligned} \tag{2}$$

where i and j index the four oscillators (one per leg: 1 =LF, 2 =RF, 3 =LH, 4 =RH), $n = 4$ is the total number of oscillators in the network, u_i and v_i are oscillator i's excitatory and inhibitory activities as in (1), $W \in \mathbb{R}^{4\times4}$ is the connecting weight matrix, W_{ij} is the entry coupling oscillator j's activity into oscillator i's dynamics, and y^i_{out} is oscillator i's output signal.

Following [14], the entries w_{ij} of W are chosen according to three principles: (i) $w_{ii} = 0$, since an oscillator does not need to influence its own dynamics; (ii) $w_{ij} > 0$ if legs i, j move in phase and $w_{ij} < 0$ if they move in anti-phase; and (iii) $|w_{ij}|$ reflects the coupling strength, set to 0.1 throughout this work. This yields the connecting weight matrices for the four gaits:

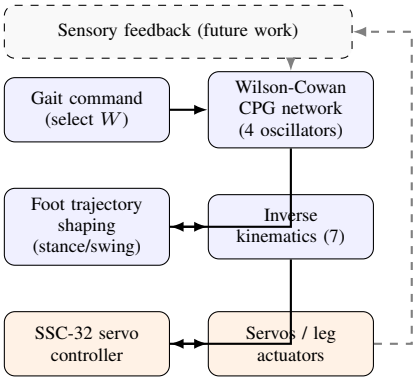


Fig. 1: High-level processing pipeline: a gait command selects the connecting weight matrix W of the Wilson-Cowan CPG network; each oscillator's output shapes a foot trajectory, converted to joint angles by inverse kinematics and sent to the SSC-32, which drives the leg actuators. The sensory-feedback path (dashed) is not implemented in the present system.

$$W_{walk} = \begin{bmatrix} 0 & -0.1 & -0.1 & -0.1 \\ -0.1 & 0 & -0.1 & -0.1 \\ -0.1 & -0.1 & 0 & -0.1 \\ -0.1 & -0.1 & -0.1 & 0 \end{bmatrix} \quad (3)$$

$$W_{trot} = \begin{bmatrix} 0 & -0.1 & -0.1 & 0.1 \\ -0.1 & 0 & 0.1 & -0.1 \\ -0.1 & 0.1 & 0 & -0.1 \\ 0.1 & -0.1 & -0.1 & 0 \end{bmatrix} \quad (4)$$

$$W_{pace} = \begin{bmatrix} 0 & -0.1 & 0.1 & -0.1 \\ -0.1 & 0 & -0.1 & 0.1 \\ 0.1 & -0.1 & 0 & -0.1 \\ -0.1 & 0.1 & -0.1 & 0 \end{bmatrix} \quad (5)$$

$$W_{gallop} = \begin{bmatrix} 0 & 0.1 & -0.1 & -0.1 \\ 0.1 & 0 & -0.1 & -0.1 \\ -0.1 & -0.1 & 0 & 0.1 \\ -0.1 & -0.1 & 0.1 & 0 \end{bmatrix} \quad (6)$$

The CPG model has a fixed topological structure; different W matrices generate rhythmic signals with the phase orders required by the walk, trot, pace and gallop gaits, without any other change to the network [14].

Fig. 1 shows the complete processing pipeline of the embedded system described by (2) and Section III: a gait command selects one of the four connecting weight matrices (3)–(6), which configures the four coupled Wilson-Cowan oscillators (one per leg); each oscillator's normalized output shapes that leg's foot trajectory (stance/swing), which is converted to the three joint angles by the inverse kinematics of (7) and sent as position commands to the SSC-32 servo controller, which drives the physical actuators. The system is currently open-loop (no sensory feedback path), which is why Section IV lists closing this loop as future work.



Fig. 2: Hardware platform: Arduino-compatible BotBoarduino (left), SSC-32 servo controller (center), and the Lynxmotion SQ3U Symmetric Quadruped robot used in the experiments (right).

TABLE II: Approximate hardware cost (USD, single unit)

Component	**Qty**	**Approx. cost**
BotBoarduino (Arduino-compatible)	1	\$45
SSC-32 servo controller	1	\$40
Hitec HS-645MG servo (coxa)	4	\$40 ea. (≈\$160)
Hitec HS-422 servo (femur/tibia)	8	\$15 ea. (≈\$120)
Chassis, hardware, cabling, battery	1	≈\$80
Total		**≈\$445**

III. Hardware Implementation

The controller was validated on a Lynxmotion SQ3U Symmetric Quadruped Walking Robot (Fig. 2, right), a commercially available research-level platform. This robot has a three-degrees-of-freedom (DOF) leg design (coxa, femur, tibia): four legs use a Hitec HS-645MG servo at the coxa joint and HS-422 servos at the femur and tibia joints, for a total of 12 DOF (4 legs $\times$ 3 DOF), consistent with the four-oscillator CPG architecture in which each Wilson-Cowan oscillator drives one leg.

The electronics consist of an Arduino-compatible board (BotBoarduino, Fig. 2, left), based on an Atmel microcontroller and designed for Lynxmotion robots, with a 5 Vdc/1.5 A regulator and up to 18 servo outputs. An SSC-32 servo controller (Fig. 2, center) drives the servomotors, providing 32 channels of 1 μs-resolution servo control with bidirectional communication with the host microcontroller. Both boards ship as the standard electronics of the SQ3U combo kit.

A. Approximate Hardware Cost

To substantiate the "low-cost" claim, Table II lists the main off-the-shelf components and their approximate current retail price (USD, single-unit, as of 2026; the original SQ3U kit is now discontinued, so prices reflect its constituent parts sold separately). The total is one to two orders of magnitude below typical FPGA development boards with comparable I/O count, and does not require the HDL toolchain or synthesis license used in our prior FPGA-based implementation [15].

B. Joint-Level Mapping: Inverse Kinematics

Each oscillator output y^i_{out} does not drive its leg's three joints directly; instead, it modulates a parametrized foot trajectory in the leg's sagittal plane, split into a stance phase (foot on the ground, moving backward relative to the body at the gait's translation speed) and a swing phase (foot lifted and moving forward), with the split point set by the gait's duty

factor. This yields a target foot position (x_f, y_f, z_f) in the leg's local frame at every control step, which is converted to the three joint angles by standard geometric inverse kinematics for a 3-DOF coxa-femur-tibia leg:

$$\begin{aligned} \theta_{coxa} &= \operatorname{atan2}(y_f,\, x_f); \\ L &= \sqrt{x_f^2 + y_f^2} - l_{coxa}; \\ D &= \sqrt{L^2 + z_f^2}; \\ \theta_{femur} &= \operatorname{atan2}(z_f, L) - \arccos\!\left(\tfrac{l_{femur}^2 + D^2 - l_{tibia}^2}{2\, l_{femur} D}\right); \\ \theta_{tibia} &= \pi - \arccos\!\left(\tfrac{l_{femur}^2 + l_{tibia}^2 - D^2}{2\, l_{femur} l_{tibia}}\right); \end{aligned} \tag{7}$$

where l_{coxa}, l_{femur} and l_{tibia} are the leg's link lengths. The resulting $(\theta_{coxa}, \theta_{femur}, \theta_{tibia})$ are the position commands sent to the corresponding SSC-32 channels for that leg.

The Wilson-Cowan oscillator network of (2) was implemented in Simulink (v. 8.5) with $T_u = T_v = 0.2$, $a = b = 5.5$, $c = 2.5$, $d = 0$, $S_u = S_v = 0$, $\mu = 0.5$ and $\alpha = 1$, following [14]. The output of each oscillator, $y_{out}^i = u_i - v_i$, was normalized to a defined interval before being sent to the servo controller.

IV. Results

Fig. 3 shows the real-time continuous-time output of the four oscillators for each gait, obtained with the *Simulink Support Package for Arduino Hardware* and a PWM output through a low-pass RC filter. For all four gaits, the generated signals are periodic and preserve the phase relationships imposed by the corresponding weight matrix: an approximately uniform phase lag for walk, diagonal pairs in phase for trot, lateral pairs in phase for pace, and fore/hind pairs in phase for gallop. These results were independently confirmed on a digital oscilloscope (Fig. 4).

These phase relationships follow directly from the sign pattern of each W matrix in (3)–(6): a negative w_{ij} drives oscillators i and j toward anti-phase, while a positive w_{ij} drives them toward in-phase locking. In W_{walk}, all off-diagonal entries are negative, so no two legs are locked in phase and the network settles into the sequential, evenly-staggered pattern characteristic of the walk gait. In W_{trot}, the diagonal pairs (LF-RH and RF-LH) have positive coupling ($w_{14} = w_{41} = w_{23} = w_{32} = 0.1$) while the remaining pairs are negative, producing the expected diagonal synchrony and an approximately anti-phase relationship between diagonal pairs. Analogous reasoning explains the lateral synchrony of pace (W_{pace}, positive coupling between same-side legs: $w_{13} = w_{31} = w_{24} = w_{42} = 0.1$) and the fore/hind synchrony of gallop (W_{gallop}, positive coupling between LF-RF and LH-RH). This confirms that gait identity is encoded entirely in the sign pattern of W, which is what allows gait switching to be implemented as a simple matrix replacement rather than a re-design of the network.

Finally, the continuous oscillatory signals were discretized into the foot-lift/foot-contact sequences of Fig. 5, which are the sequences ultimately sent to the servo controller. For all four gaits, the discretized patterns preserve the phase order imposed by the connecting weight matrix used to generate them, confirming that the embedded system correctly translates the continuous CPG dynamics into the discrete commands required to actuate the robot in real time.

A. Physical Locomotion

Beyond the oscillator- and joint-command-level validation above, the four gaits were physically executed on the Lynxmotion SQ3U platform: the robot's legs reproduced each gait's commanded motion in real time while tethered to the host computer, and forward locomotion was additionally recorded on video. Fig. 6 shows the real-time test setup, with the robot's legs captured in motion. This confirms that the commands generated by the embedded CPG are sufficient to produce coordinated leg motion on the physical platform, and not only the phase-correct signals shown in Figs. 3–5.

B. Real-Time Operation and Limitations

In this work, "real time" refers to the embedded system generating and transmitting each leg's joint commands within a fixed control cycle, without accumulating latency or missing updates, so that the commanded trajectory keeps pace with the oscillator dynamics rather than being computed off-line and replayed. We did not instrument the deployed BotBoarduino sketch to log its actual loop period or jitter, so we cannot report a measured control frequency or a quantitative comparison of execution time, frequency accuracy, or joint-tracking error (e.g., RMSE) against our earlier FPGA-based implementation [15]; the values $T_u = T_v = 0.2$ in Section II are the model's own dimensionless time constants and are not a measurement of the hardware loop rate. Qualitatively, the FPGA implementation provides deterministic, fully parallel updates of all oscillators with sub-microsecond timing resolution, whereas the microcontroller in this work updates the network sequentially and is subject to loop-time jitter from the underlying Arduino runtime; quantifying this gap, together with the resulting joint-tracking error relative to the CPG reference, is left as future work and would require timestamped logging of both the CPG output and the physical joint angles, which was not available in the present deployment.

V. Conclusion

We presented a low-cost and fully reconfigurable embedded system for real-time, multi-gait locomotion in legged robots, based on a Central Pattern Generator composed of four coupled Wilson-Cowan neural oscillators running on a single Arduino-based microcontroller. By simply replacing the connecting weight matrix – without modifying the underlying neural architecture or re-programming the embedded system – the same controller reproduced the four canonical quadruped gaits: walk, trot, pace and gallop. Real-time hardware validation, using a Simulink-Arduino co-simulation and verified on a digital oscilloscope, showed that the generated signals preserve the expected phase relationships, that the resulting discretized patterns are correctly delivered to the servo controller, and

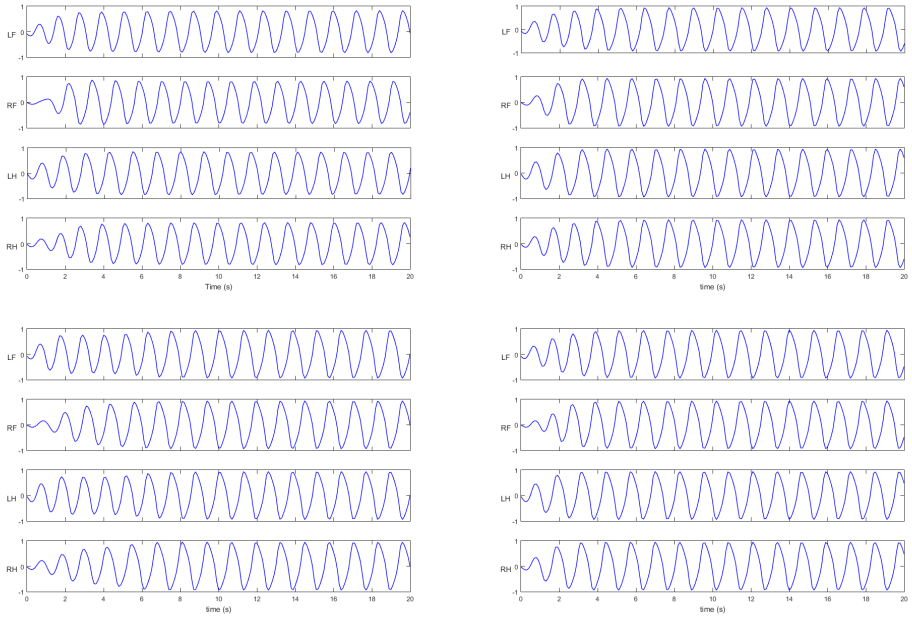


Fig. 3: Real-time continuous-time oscillator outputs for the four gaits (top-left: walk, top-right: trot, bottom-left: pace, bottom-right: gallop). L, R, F, H denote left, right, front, hind; x-axis: seconds, y-axis: radians.

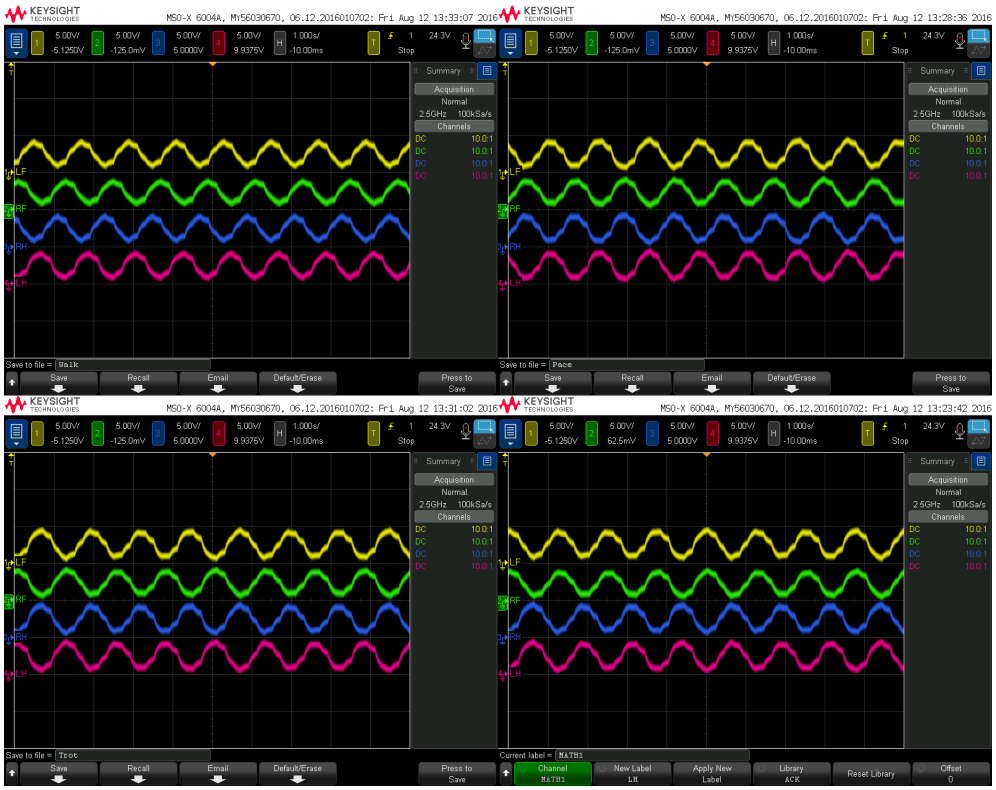


Fig. 4: Oscilloscope screenshots confirming the phase relationships for walk (top-left), trot (top-right), pace (bottom-left) and gallop (bottom-right).

that the corresponding joint commands produce physical locomotion of the robot for each gait. The network's topology is scalable in the sense that it can be extended to additional oscillators (e.g., for a hexapod) by adding rows and columns to W following the same three design principles; we have not, however, empirically validated this beyond the four-leg



Fig. 5: Discretized gait patterns sent to the servo controller (top-left: walk, top-right: trot, bottom-left: pace, bottom-right: gallop).

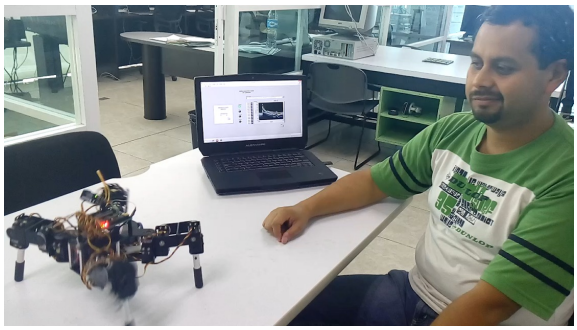


Fig. 6: Real-time hardware-in-the-loop test setup: the quadruped robot executing a gait (legs shown in motion) while tethered to the host laptop running the Simulink external-mode interface used to command and monitor the CPG.

case studied here. These results support CPG-based controllers as a lightweight, reconfigurable alternative to trajectory-based locomotion control in resource-constrained legged robots.

Future work will focus on: closing the loop with sensory feedback for terrain adaptation; enabling smooth continuous gait transitions; instrumenting the embedded system to log control-loop timing and physical joint angles, so that execution time, achievable oscillation frequency, and joint-tracking error can be quantified and compared against the FPGA implementation [15]; evaluating the network's robustness to external perturbations that disturb its phase or oscillatory state, using metrics such as phase-resynchronization time and post-perturbation gait stability; and quantifying energetic efficiency across gaits on a fully actuated platform.

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